Anirban Majumdar

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Professional Summary

Automated Reasoning researcher with 6+ years of experience in *verification*, *automata learning*, and *decision processes*. Demonstrated success in developing novel algorithms and tools in symbolic AI and automated reasoning. Experienced in Python, with a strong publication record in top venues such as STACS, CONCUR, ATVA, FSTTCS, etc.

Technical Skills

- **Languages:** Python, Haskell, Matlab, HTML.
- **Formal Tools:** PRISM (conceptual), TChecker (familiar), Z3 (familiar), XCos (familiar), NuSMV (conceptual), Uppaal (conceptual).
- **Concepts:** Reinforcement Learning, Automated Reasoning, Temporal logics (LTL, CTL), Automata theory, (Partially Observable-) MDPs, Real-time systems.
- **Other:** Git, Scientific writing, Research prototyping.

Experience

Postdoctoral Researcher December 2025 – Present
Tata Institute of Fundamental Research (TIFR), India

Independent Researcher December 2024 – November 2025
Focused on symbolic reasoning in decision making under uncertainty and partial observability (POMDPs), reinforcement learning with timed specifications, learning algorithms for counter automata, with tool development.

Postdoctoral Researcher December 2021 – November 2024
Université Libre de Bruxelles (ULB), Belgium
Worked on learning of timed systems and symbolic analysis of MDPs. Developed tools for (active and passive) learning of timed systems and contributed to publications in top-tier conferences.

Featured Projects (Tools)

tLsep – Greybox Learning of ERA-recognizable languages
Python implementation of an active learning algorithm for event-recording automata.
github.com/mukherjee-sayan/ERA-greybox-learn

LEAP – Passive Learning of ERA-recognizable languages
Python implementation of a passive learning algorithm for event-recording automata.
github.com/anirban11/leap

Robbins_MDP – Solving Robbins' Problem using MDPs
Algorithmic exploration of optimal stopping problems using symbolic methods.
github.com/anirban11/Robbins_MDP

Education

Ph.D. in Computer Science 2018–2021
ENS Paris-Saclay, France

M.Sc. in Computer Science 2016–2018
Chennai Mathematical Institute, India

B.Sc. in Mathematics and Computer Science 2013–2016
Chennai Mathematical Institute, India

Publications (Selected) [DBLP] [Scholar]

1. **About Time: Model-free Reinforcement Learning with Timed Reward Machines**, *accepted at* IJCAI 2026.
2. **Synthesizing POMDP Policies: Sampling Meets Model-checking via Learning**, *accepted at* CAV 2026.
3. **Scalable Learning of One-Counter Automata via State-Merging Algorithms**, FSTTCS 2025
4. **Learning Event-recording Automata Passively**, ATVA 2025
5. **Greybox Learning of Languages Recognizable by ERA**, ATVA 2024
6. **Algorithms for Robbins' Problem Using MDPs**, Principles of Verification: Cycling the Probabilistic Landscape, 2024
7. **Bi-objective Lexicographic Optimization in MDPs**, ATVA 2023
8. **Synthesizing Safe Coalition Strategies**, FSTTCS 2020
9. **Concurrent Parameterized Games**, FSTTCS 2019
10. **Reconfiguration and Message Losses in Parameterized Broadcast Networks**, CONCUR 2019
11. **Width of Non-deterministic Automata**, STACS 2018

Theses

Ph.D. Thesis: *Verification and Synthesis of Parameterized Concurrent Systems*
Supervisors: Patricia Bouyer-Decitre, Nathalie Bertrand

M.Sc. Thesis: *Playing with Repetitions in Data Words*
Supervisor: M. Praveen

References

Available on request.